

DEPARTMENT OF MATHEMATICS

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Welcome Message

The Department of Mathematics at the Royal University of Phnom Penh is the oldest and largest mathematics department in Cambodia. Founded in 1960, we have a proud history of excellence in undergraduate training, particularly in mathematics, and have prepared generations of mathematics teachers who serve across the country. Today our department offers rigorous programs in pure and applied mathematics and statistics supported by a dedicated faculty engaged in teaching, research, and community service. We emphasize strong foundations in theory and applications with data and computational tools, and opportunities for research, internships, and national and international collaboration.

The Department collaborates with government agencies, foreign embassies, schools, industry partners, and universities in the region and beyond to deliver joint research, training workshops, exchange programs, and outreach activities. These partnerships create opportunities for internships, scholarships, student and faculty exchanges, and conference participation that broaden experience and enhance career prospects.

Building on this strong support and collaboration, the Department will launch an international dual-degree program in 2028. The program will offer majors aligned with Cambodia's digital development priorities, including Mathematics, Statistics, Mathematics–Economics, Engineering–Mathematics, Mathematics Education, and Mathematics–Informatics. Through this initiative we aim to prepare graduates with the analytical, computational, and interdisciplinary skills needed to excel as educators, researchers, analysts, and innovators in a rapidly evolving, data-driven world.

We warmly invite universities, research institutes, industry partners, foreign governments, and organizations to collaborate with us on shared priorities, including joint research, student and faculty exchanges, curriculum development, and capacity building, to deepen mathematical knowledge, strengthen professional skills, and advance Cambodia's educational and scientific development. For more details, please see our faculty and staff profiles or contact the Department office. Thank you!

About the Department!

The Department of Mathematics at the Royal University of Phnom Penh (RUPP) is the oldest and largest mathematics department in Cambodia. Established in 1960, the Department has played a central role in undergraduate mathematics education nationwide and remains a primary source of trained mathematics teachers for the country's high schools. The Department comprises 21 academic staff, including six PhD holders in mathematics and related fields, and is actively expanding faculty research qualifications and postgraduate supervision capacity to increase research output improve the quality and relevance of theses and publications. We offer comprehensive undergraduate (Bachelor) program in Mathematics

Statistics and have steadily expanded our postgraduate offerings. In 2007 the Department launched a Master's programme in collaboration with partner universities across the region (India, the Philippines, Vietnam, Laos, Myanmar), East Asia and Oceania (South Korea, Australia), Europe (France, Germany, Spain, Sweden), and the USA. Building on this foundation, in 2019, together with Linköping University and Stockholm University (Sweden) and regional partners, we developed an internationally oriented MSc (coursework and dissertation) designed to prepare students for PhD study. In 2022 we established a Doctoral Program in Mathematics to further strengthen research capacity and supervision. Aligned with RUPP's vision to become Cambodia's flagship university in teaching, research, and community service, the Department is committed to delivering high-quality education, expanding postgraduate training, fostering international collaboration, and contributing to Cambodia's scientific and socio-economic development.

Vision

We envision becoming a regionally recognized Department of Mathematics in higher education, renowned for excellence in teaching, research, and publication, and for contributing to Cambodia's social and economic development.

Mission

The Department of Mathematics is committed to educating graduates who excel in mathematics and statistics and apply their knowledge for societal benefit. Our core missions are to:

- Foster high-quality research and scholarly publication that advances knowledge and supports national development.
- Provide research, expertise, and services to the public and private sectors.
- Continuously expand and adapt our programmes to meet emerging needs in Cambodia and the region.

Goals

The Department has set the following goals:

- Achieve accreditation from the Accreditation Committee of Cambodia (ACC) and the ASEAN University Network (AUN).
- Attain excellence in teaching and research across mathematics, statistics, and multidisciplinary fields.
- Provide high-quality research and services to public and private sectors in pure and applied mathematics and statistics.
- Regularly update curricula and expand student and staff exchange programmes with regional partners.

Program Specifications

The Department currently delivers in-house undergraduate and graduate programs in Mathematics, including BSc degrees, an MSc program, and PhD studies. These programs emphasize strong theoretical foundations, applied skills, and opportunities for research, internships, and professional development.

In response to national priorities, we plan to launch an international bachelor's program (dual-degree) in 2028 with majors aligned to the Royal Government of Cambodia's development needs, including Engineering–Mathematics; Mathematics–Informatics (Computational Mathematics); Mathematics–Economics; Statistics and Data Science; Mathematics Education; and Applied/Computational Modeling. These majors are designed to equip students with analytical, computational, and interdisciplinary skills for careers in education, industry, government, and research. Further details on curricula, admission criteria, and program partnerships will be published as plans are finalized.

BSc in Mathematics

Program Goals / Objectives

The Bachelor of Science in Mathematics provides strong foundations in pure and applied mathematics, mathematical modelling, and programming skills. The programme prepares graduates for careers and further study in natural sciences, engineering, social sciences, finance, and data-driven industries.

The program aims to

- develop leaders in mathematics and applied mathematics.
- educate professional mathematicians, including secondary-school teachers.
- prepare graduates to contribute to research institutes in science and engineering.
- strengthen graduates' ability to apply mathematics across disciplines.
- introduce mathematical foundations for machine learning and artificial intelligence.

Program Learning Outcomes (PLOs)

After graduation, students will be able to:

a. Knowledge

- PLO1: Describe natural and social phenomena using mathematical models and statistical concepts.
- PLO2: Explain mathematical results and implement them using appropriate programming languages.
- PLO3: Demonstrate readiness for postgraduate study in mathematics or related fields.
- PLO4: Apply programming (e.g., Python, MATLAB) to solve problems in economics, engineering, and applied contexts.

b. Cognitive skills

- PLO5: Analyse and solve mathematical problems using rigorous reasoning and practical methods.
- PLO6: Apply mathematical tools and data-analysis techniques across diverse fields.
- PLO7: Conduct research, present findings, and formulate critical questions supported by analysis and discussion.
- PLO8: Prepare clear technical materials with appropriate references and critical appraisal.

c. Interpersonal skills and responsibility

- PLO9: Apply mathematical methods ethically and responsibly, integrating ICT to address societal needs.
- PLO10: Work independently and collaboratively, demonstrating accountability and commitment to lifelong learning.

d. Numerical skills, information technology and communication

- PLO11: Use computational tools to model, analyse, and solve mathematical problems.
- PLO12: Select and communicate appropriate mathematical and statistical methods for analysis, prediction, and presentation.

e. Practical / psychomotor skills

- PLO13: Use ICT tools for calculation, numerical analysis, approximation, and informed decision-making.

Curriculum Structure

The Bachelor of Science in Mathematics or Statistics is a four-year, in-house undergraduate program that builds strong foundations in pure and applied mathematics and in applied statistics. The first two years cover common foundation courses; students choose their major (Mathematics or Statistics) at the end of Year 2. The program combines core theories and applications and includes a final-year capstone project to prepare graduates for careers in teaching, industry, finance, IT, and for postgraduate study.

Year I					
Semester I			Semester II		
Code	Course Title	Credit#	Code	Course Title	Credit #
	English Language I	2(1.2.0)		English Language II	2(1.2.0)
FGKL02	Khmer Culture and Civilization	2(2.0.0)	FGPP01	Introduction To Philosophy	2(2.0.0)
FGMA01	General Mathematics	2(2.0.0)	FGMA02	Statistics	2(2.0.0)
FGCH01	General Chemistry	2(2.0.0)	FGBI01	General Biology	2(2.0.0)
FGPH01	General Physic	2(2.0.0)	FGCS01	Computer Applications	1(0.5.1.0)
FGPS01	General Phycology	2(2.0.0)	FGGE01	Demographics	2(2.0.0)
FEMA01	General Algebra I	3(3.0.0)	FEMA02	General Algebra II	3(3.0.0)
FEMA03	General Analysis I	3(3.0.0)	FEMA04	General Analysis II	3(3.0.0)
	Total	16		Total	17

Year II	
Semester I	Semester II

Code	Course Title	Credit#	Code	Course Title	Credit #
	English Language III	4(2.4.0)		English Language IV	4(2.4.0)
SMA201	Programming Language (Python)I	2(1.2.0)	SMA206	Programming Language (Python)II	2(1.2.0)
SMA202	General Analysis III	5(4.2.0)	SMA207	General Analysis IV	5(4.2.0)
SMA203	General Algebra III	4(3.2.0)	SMA208	General Algebra IV	4(3.2.0)
SMA204	Introduction to StatisticsI	2(2.0.0)	SMA209	Introduction to Statistics II	2(2.0.0)
SMA205	Linear Algebra I	3(2.2.0)	SMA210	Linear Algebra II	3(2.2.0)
	Total	20		Total	20

Year III					
Semester I			Semester II		
Code	Course Title	Credit#	Code	Course Title	Credit #
	English Language V	4(2.4.0)		English Language VI	4(2.4.0)
SMA301	Topology I	5(4.2.0)	SMA306	Topology II	5(4.2.0)
SMA302	Advanced Analysis I	3(2.2.0)	SMA307	Advanced Analysis II	3(2.2.0)
SMA302	Geometry I	3(2.2.0)	SMA308	Geometry II	3(2.2.0)
SMA304	Probability I	2(1.2.0)	SMA309	Probability II	2(1.2.0)
SMA305	Operations Research I	2(1.2.0)	SMA310	Operations Research II	2(1.2.0)
	Total	20		Total	20

Year IV					
Semester I			Semester II		
Code	Course Title	Credit#	Code	Course Title	Credit #
SMA401	Applied statistics I	3(2.2.0)	SMA406	Applied statistics II	3(2.2.0)
SMA402	Complex Analysis	5(4.2.0)	SMA407	Measure Theory	5(4.2.0)
SMA403	Differential Calculus I	5(4.2.0)	SMA408	Differential Calculus II	5(4.2.0)
SMA404	Advanced Algebra I	5(4.2.0)	SMA409	Advanced Algebra II	5(4.2.0)
SMA405	Numerical Analysis	3(2.2.0)	SMA410	Financial Mathematics	3(2.2.0)
				Or Thesis	
	Total	21		Total	21

Explanation:

The credit description of each course shall be defined as follows:

- Course means subject to be studies per semester or term.
- The code a(b-c-d) indicates the number of credits and weekly study load.
In this example, 'a' =number of credits, 'b'=number of lecture hours, 'c'= number of tutorial or practical hours, 'd'= number of field work/Internship hours.

- Lecture (Class or Exercise): 15 hours/1 Credit
- Experiment or Practical Work: 30 hours/1 Credit
- Field Work/Internship: 45 hours/ 1 Credit

BSc in Statistics

Program Goals / Objectives

The Bachelor of Science in Statistics is a four-year programme designed to provide solid grounding in statistical theory, applied statistics, data analysis, and computational methods. The programme develops students' ability to design studies, analyse data, interpret results, and apply statistical methods across natural sciences, engineering, economics, public policy, health, and industry. Graduates are prepared for professional roles, further graduate study, and contributions to research and evidence-based decision making.

The program aims to

- Develop competent statisticians and data analysts for education, industry, government, and research.
- Prepare graduates to pursue postgraduate studies in statistics, data science, and related fields.
- Equip students with applied statistical skills for research institutes and multidisciplinary projects.
- Strengthen programming and computational competence for real-world data analysis (e.g., Python, R, MATLAB).
- Introduce foundations of statistical learning, machine learning, and data science.

Program Learning Outcomes (PLOs)

After completing the BSc in Statistics, graduates will be able to:

a. Knowledge

- PLO1: Describe and model natural and social phenomena using statistical concepts and probability theory.
- PLO2: Explain statistical methods and their assumptions, and implement them using suitable programming languages (e.g., R, Python, MATLAB).
- PLO3: Demonstrate readiness for postgraduate study in statistics, data science, or related disciplines.
- PLO4: Apply statistical methods to solve problems in economics, engineering, health, and other applied domains.

b. Cognitive Skills

- PLO5: Formulate appropriate statistical questions, select suitable methods, and produce valid conclusions from data.
- PLO6: Use statistical tools and data-analysis techniques to explore, model, and interpret complex datasets.
- PLO7: Design and conduct applied research projects, including hypothesis formulation, experimental/ survey design, analysis, and presentation of results.
- PLO8: Critically review statistical literature and prepare technical reports with clear argumentation and references.

c. Interpersonal Skills & Responsibility

- PLO9: Apply statistical methods ethically and responsibly, considering data privacy, reproducibility, and societal impact.
- PLO10: Work effectively both independently and in teams, demonstrating professional responsibility and commitment to lifelong learning.

d. Numerical, IT & Communication Skills

- PLO11: Use computational and statistical software tools to perform data cleaning, analysis, visualization, and modelling.
- PLO12: Select and communicate appropriate statistical methods and results for decision making, policy, and stakeholder presentations.

e. Practical / Psychomotor Skills

- PLO13: Implement data-analysis workflows using ICT tools, perform numerical computations, and make evidence-based recommendations.

Curriculum Structure

Year I					
Semester I			Semester II		
Code	Course Title	Credit#	Code	Course Title	Credit #
	English Language I	2(1.2.0)		English Language II	2(1.2.0)
FGKL02	Khmer Culture and Civilization	2(2.0.0)	FGPP01	Introduction To Philosophy	2(2.0.0)

FGMA01	General Mathematics	2(2.0.0)	FGMA02	Statistics	2(2.0.0)
FGCH01	General Chemistry	2(2.0.0)	FGBI01	General Biology	2(2.0.0)
FGPH01	General Physic	2(2.0.0)	FGCS01	Computer Applications	1(0.5.1.0)
FGPS01	General Phycology	2(2.0.0)	FGGE01	Demographics	2(2.0.0)
FEMA01	General Algebra I	3(3.0.0)	FEMA02	General Algebra II	3(3.0.0)
FEMA03	General Analysis I	3(3.0.0)	FEMA04	General Analysis II	3(3.0.0)
	Total	16		Total	17

Year II					
Semester I			Semester II		
Code	Course Title	Credit#	Code	Course Title	Credit #
	English Language III	4(2.4.0)		English Language IV	4(2.4.0)
SMA201	ProgrammingLanguage (Python)I	2(1.2.0)	SMA206	ProgrammingLanguage (Python)II	2(1.2.0)
SMA202	General Analysis III	5(4.2.0)	SMA207	General Analysis IV	5(4.2.0)
SMA203	General Algebra III	4(3.2.0)	SMA208	General Algebra IV	4(3.2.0)
SMA204	Introduction to Statistics I	2(2.0.0)	SMA209	Introduction to Statistics II	2(2.0.0)
SMA205	Linear Algebra I	3(2.2.0)	SMA210	Linear Algebra II	3(2.2.0)
	Total	20		Total	20

YEAR III					
Semester I			Semester II		
Code	Course Title	Credit#	Code	Course Title	Credit #
	English Language V	4(2.4.0)		English Language VI	4(2.4.0)
SST301	Theory of Statistics I	3(2.2.0)	SST306	Theory of Statistics II	3(2.2.0)
SST302	Introduction to Insurance	3(2.2.0)	SST307	Mathematics for Life Insurance	3(2.2.0)
SST311	R Programming	3(2.2.0)	SST406	Sampling Techniques	3(2.2.0)
SST304	Probability I	2(1.2.0)	SST309	Probability II	2(1.2.0)
SST305	Operations Research I	2(1.2.0)	SST310	Operations Research II	2(1.2.0)
	Total	20		Total	20

YEAR IV					
Semester I			Semester II		
Code	Course Title	Credit#	Code	Course Title	Credit #
SST401	Linear Regression I	5(4.2.0)	SST406	Linear Regression I	5(4.2.0)
SST402	Stochastic Process I	3(2.2.0)	SST407	Stochastic Process I	3(2.2.0)

SST411	Numerical Analysis for Statistics I	5(4.2.0)	SST412	Numerical Analysis for Statistics II	5(4.2.0)
SST404	Multivariate Analysis I	3(2.2.0)	SST409	Multivariate Analysis II	5(4.2.0)
SST308	Research Methodology	3(2.2.0)	SST410	Financial Mathematics	3(2.2.0)
				Or Thesis	
	Total	21		Total	21

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The credit description of each course shall be defined as follows:

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- Lecture (Class or Exercise): 15 hours/1 Credit
- Experiment or Practical Work: 30 hours/1 Credit
- Field Work/Internship: 45 hours/ 1 Credit

Admission

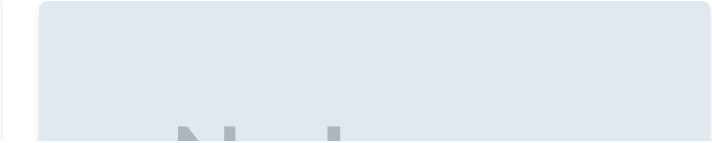
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Contact Details

-  Address
N/A
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N/A
-  Email
N/A

All Program

| Bachelor Degree



About RUPP

The Royal University of Phnom Penh is Cambodia's oldest and largest public university, offering degrees in science, technology, humanities, and social sciences.

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